# The second law

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain the direction of heat transfer and the limits of energy conversion.

**SUCCESS CRITERIA - I can...**

1. Predict spontaneous net heat flow.
2. Explain why moving heat from cold to hot needs an additional process.
3. Distinguish energy conservation from process direction.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “spontaneous” and “heat” to explain your answer. Sentence starters are optional.

**spontaneous**: able to occur without the specified external work input

**heat**: energy transferred because of a temperature difference

**work**: energy transferred by a force along a displacement

## 01 Warm-up: bring back an earlier idea

A **0.3 kg** metal cools from **91** to **31°C** while **0.25 kg** water warms from **11** to **31°C**.
1. Use water c=**4200**; neglect cup/losses.
2. **Find metal c**.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $Q_{hot} = Q_{cold} + W_{in}$.
Refrigerator cycle: heat amounts are positive magnitudes. Work input moves heat from cold to hot; spontaneous flow is hot to cold.

**Worked reasoning:** Hot and cold blocks in contact exchange energy until thermal equilibrium. Total energy can be conserved while the temperature difference and ability to drive a heat engine decrease.

**Your turn:** Draw before-and-after temperature states and the spontaneous heat-flow direction.

**Sentence starter:** In my work, I used … to show … . I labeled it … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

A **50°C** block touches a **10°C** block. **Draw net heat flow** and **predict their trend toward equilibrium**.

**Sentence starter:** I predict … . I used … to make this prediction because … .

50°C

10°C

Each outline represents the labeled sample or container.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 04 Apply the physics

During one modeled refrigerator cycle, **200 J** is removed from the cold compartment using **50 J** of electrical work.
1. **Draw the energy flows**.
2. **Find the energy released** to the room and the coefficient of performance, $Q_c/W$.
3. **Explain why COP** is not heat-engine efficiency.

**Sentence starter:** For this answer, I treated ... as true. I found ... . This supports ... because ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

A proposed machine takes heat from one reservoir and turns all of it into work in a repeating cycle. What thermodynamic limitation is missing?

**Sentence starter:** For this answer, I treated ... as true. I found ... . This supports ... because ... .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A **0.4 kg** object moving at **6 m/s** stops through friction over **0.8 m**. **Find energy transformed** and average opposing-force magnitude.
**Show the energy changes** and transfers; **label units**.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

| Initial | Transfers | Final |
|---|---|---|
| | | |

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Two bodies exchange heat in an isolated system and reach the same temperature. Is the total energy lost? **What happens** to the ability to extract work from their temperature difference?

**Sentence starter:** Both ... . They differ in ... because ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A refrigerator removes **200 J** from a cold space while receiving **80 J** electrical work. **Find heat delivered** to the room and **explain why it** is larger than **200 J**.

**Sentence starter:** Both ... . They differ in ... because ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3